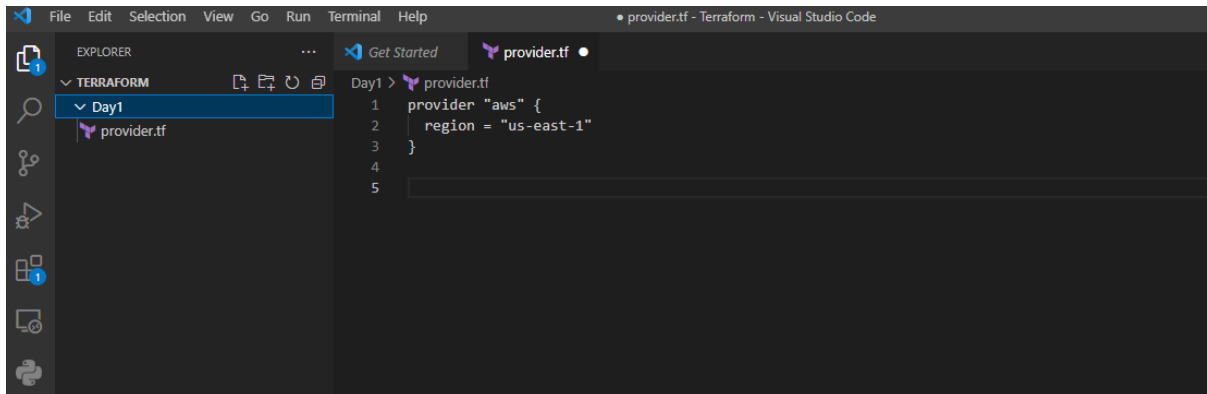
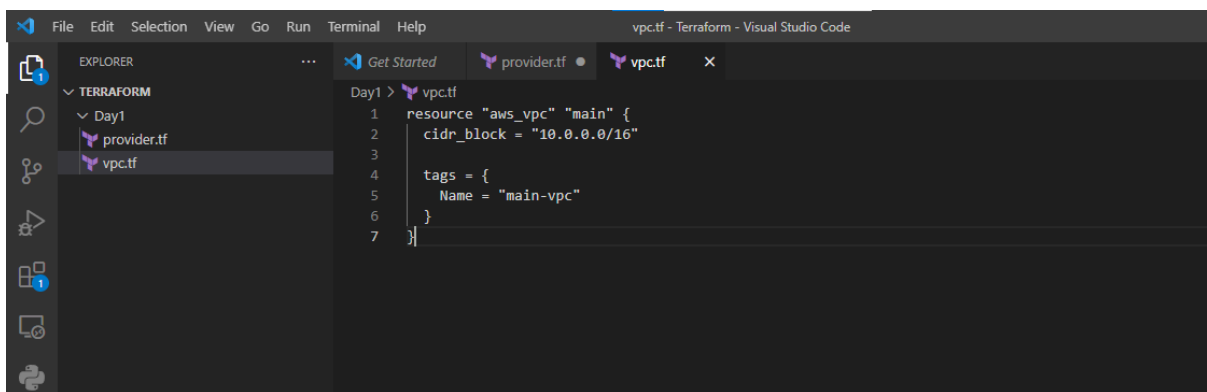


Terraform_Lab1

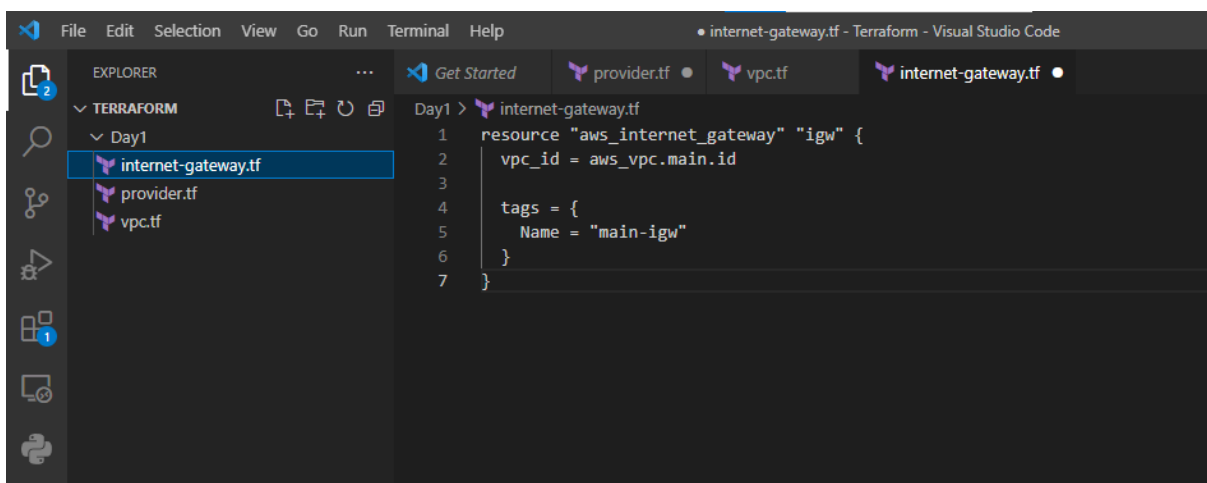


Networking

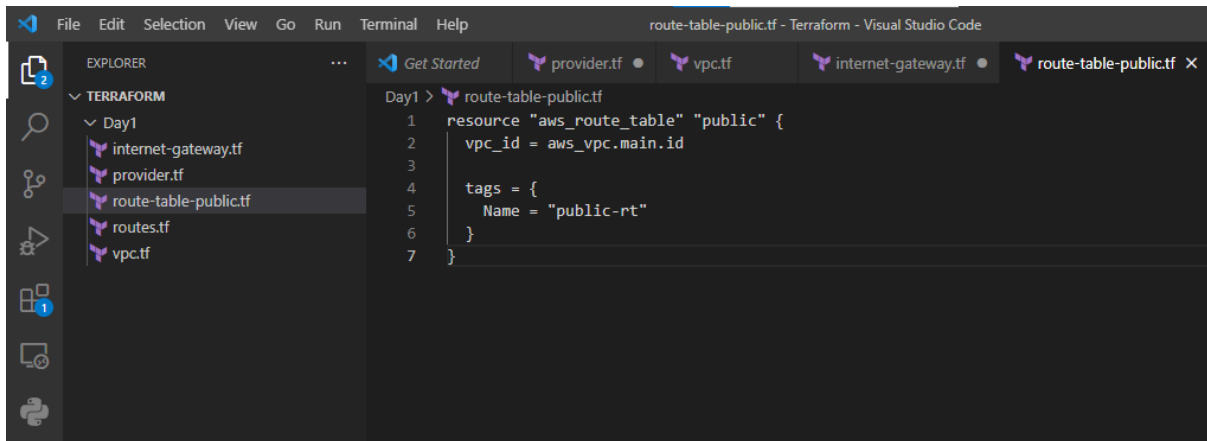
1- create vpc



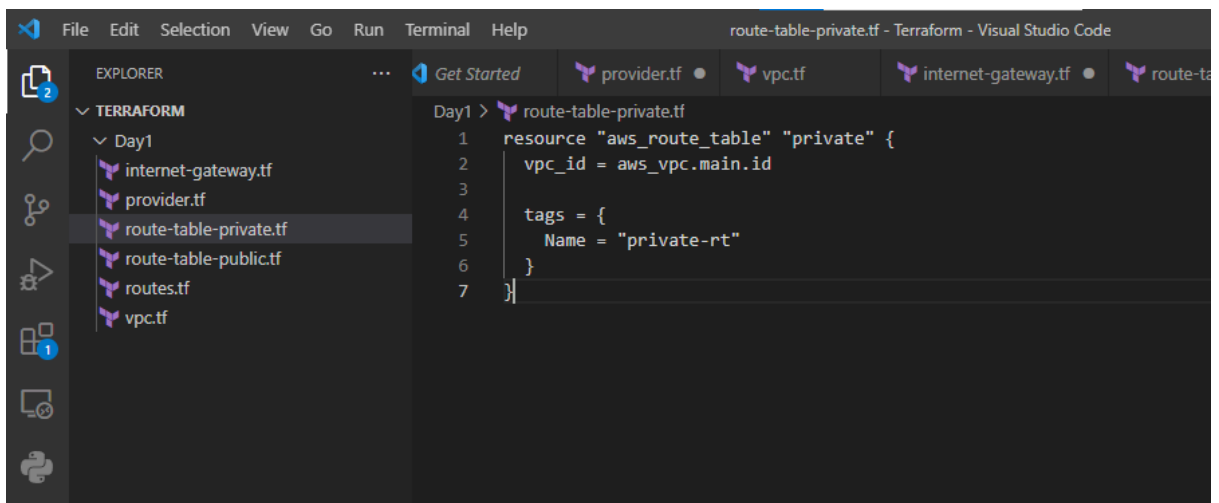
2- create internet gateway



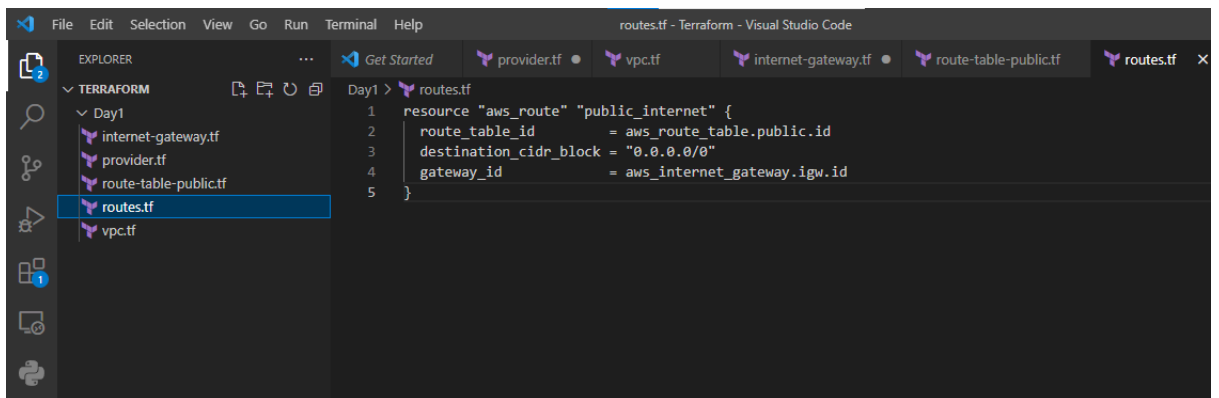
3- create public route table



4- create private route table



5- create public route



6- attach public route table to subnets

```
File Edit Selection View Go Run Terminal Help
subnet-private.tf - Terraform - Visual Studio Code

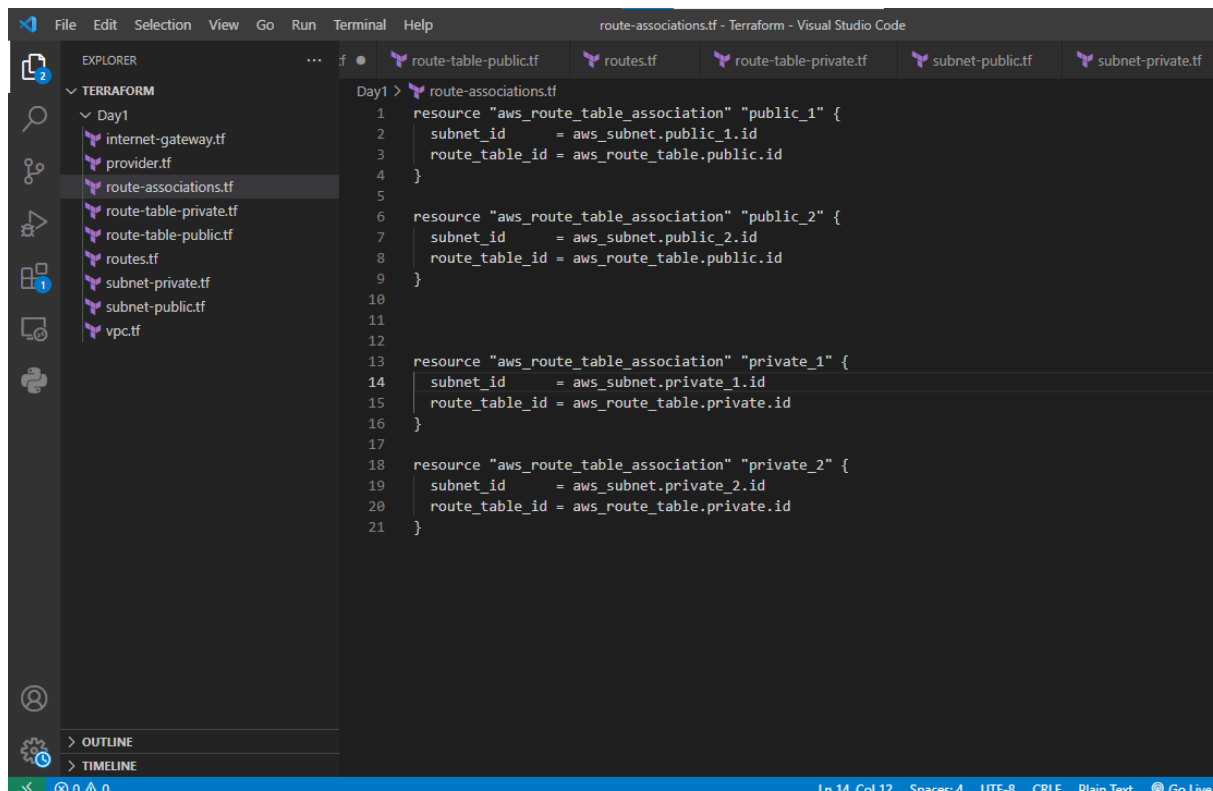
EXPLORER
TERRAFORM
  Day1
    internet-gateway.tf
    provider.tf
    route-table-private.tf
    route-table-public.tf
    routes.tf
    subnet-private.tf
    subnet-public.tf
    vpc.tf

Day1 > subnet-private.tf
1 resource "aws_subnet" "private_1" {
2   vpc_id      = aws_vpc.main.id
3   cidr_block  = "10.0.3.0/24"
4   availability_zone = "us-east-1a"
5
6   tags = {
7     Name = "private-subnet-1"
8   }
9 }
10
11 resource "aws_subnet" "private_2" {
12   vpc_id      = aws_vpc.main.id
13   cidr_block  = "10.0.4.0/24"
14   availability_zone = "us-east-1b"
15
16   tags = {
17     Name = "private-subnet-2"
18   }
19 }
```

```
File Edit Selection View Go Run Terminal Help
subnet-public.tf - Terraform - Visual Studio Code

EXPLORER
TERRAFORM
  Day1
    internet-gateway.tf
    provider.tf
    route-table-private.tf
    route-table-public.tf
    routes.tf
    subnet-private.tf
    subnet-public.tf
    vpc.tf

Day1 > subnet-public.tf
1 resource "aws_subnet" "public_1" {
2   vpc_id      = aws_vpc.main.id
3   cidr_block  = "10.0.1.0/24"
4   availability_zone = "us-east-1a"
5   map_public_ip_on_launch = true
6
7   tags = {
8     Name = "public-subnet-1"
9   }
10 }
11
12 resource "aws_subnet" "public_2" {
13   vpc_id      = aws_vpc.main.id
14   cidr_block  = "10.0.2.0/24"
15   availability_zone = "us-east-1b"
16   map_public_ip_on_launch = true
17
18   tags = {
19     Name = "public-subnet-2"
20   }
21 }
```



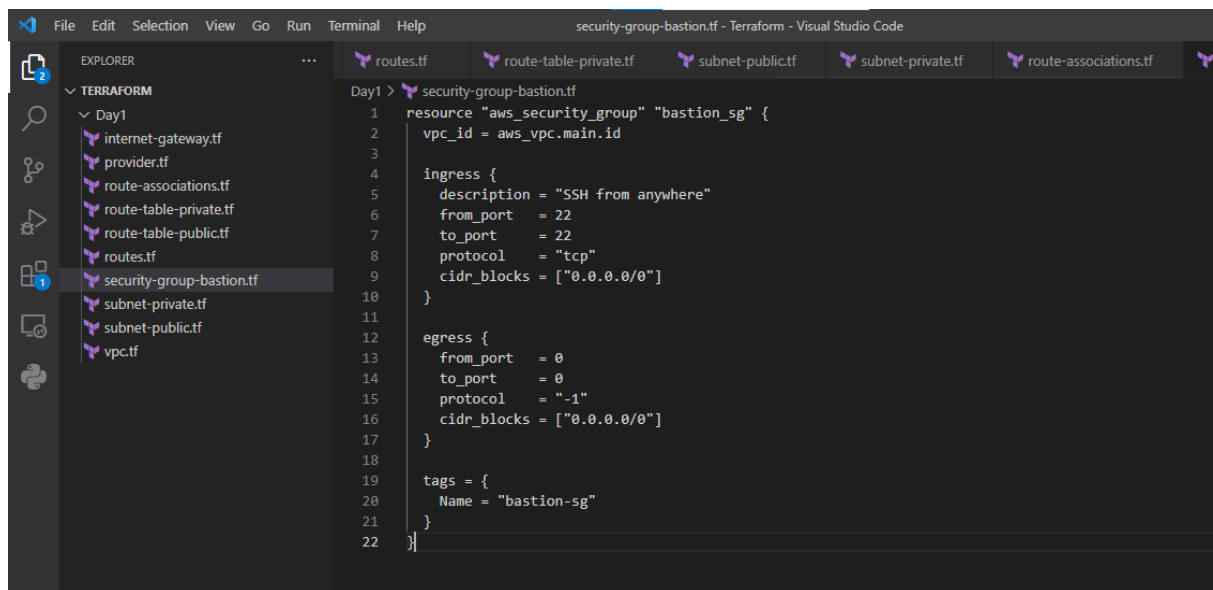
```
File Edit Selection View Go Run Terminal Help route-associations.tf - Terraform - Visual Studio Code

EXPLORER
TERRAFORM
  Day1
    internet-gateway.tf
    provider.tf
    route-associations.tf
    route-table-private.tf
    route-table-public.tf
    routes.tf
    subnet-private.tf
    subnet-public.tf
    vpc.tf

Day1 > route-associations.tf
1 resource "aws_route_table_association" "public_1" {
2   subnet_id = aws_subnet.public_1.id
3   route_table_id = aws_route_table.public.id
4 }
5
6 resource "aws_route_table_association" "public_2" {
7   subnet_id = aws_subnet.public_2.id
8   route_table_id = aws_route_table.public.id
9 }
10
11
12
13 resource "aws_route_table_association" "private_1" {
14   subnet_id = aws_subnet.private_1.id
15   route_table_id = aws_route_table.private.id
16 }
17
18 resource "aws_route_table_association" "private_2" {
19   subnet_id = aws_subnet.private_2.id
20   route_table_id = aws_route_table.private.id
21 }
22 }
```

Compute

7- create security group which allow ssh from 0.0.0.0/0



```
File Edit Selection View Go Run Terminal Help security-group-bastion.tf - Terraform - Visual Studio Code

EXPLORER
TERRAFORM
  Day1
    internet-gateway.tf
    provider.tf
    route-associations.tf
    route-table-private.tf
    route-table-public.tf
    routes.tf
    security-group-bastion.tf
    subnet-private.tf
    subnet-public.tf
    vpc.tf

Day1 > security-group-bastion.tf
1 resource "aws_security_group" "bastion_sg" {
2   vpc_id = aws_vpc.main.id
3
4   ingress {
5     description = "SSH from anywhere"
6     from_port = 22
7     to_port = 22
8     protocol = "tcp"
9     cidr_blocks = ["0.0.0.0/0"]
10  }
11
12  egress {
13    from_port = 0
14    to_port = 0
15    protocol = "-1"
16    cidr_blocks = ["0.0.0.0/0"]
17  }
18
19  tags = {
20    Name = "bastion-sg"
21  }
22 }
```

8- create security group that allow ssh and port 3000 from vpc cidr only

The screenshot shows the Visual Studio Code interface with a Terraform project. The Explorer panel on the left lists files under a 'TERRAFORM' directory, including 'Day1' and various .tf files. The 'security-group-app.tf' file is selected and its content is displayed in the main editor. The code defines an AWS security group resource named 'app_sg' with two ingress rules and one egress rule. The first ingress rule allows SSH (port 22) from the VPC. The second ingress rule allows traffic on port 3000 from the VPC. The egress rule allows all traffic (port 0) to the internet (protocol -1).

```
1 resource "aws_security_group" "app_sg" {
2   vpc_id = aws_vpc.main.id
3
4   ingress {
5     description = "SSH from VPC only"
6     from_port   = 22
7     to_port     = 22
8     protocol    = "tcp"
9     cidr_blocks = [aws_vpc.main.cidr_block]
10  }
11
12  ingress {
13    description = "App port 3000"
14    from_port   = 3000
15    to_port     = 3000
16    protocol    = "tcp"
17    cidr_blocks = [aws_vpc.main.cidr_block]
18  }
19
20  egress {
21    from_port   = 0
22    to_port     = 0
23    protocol    = "-1"
24    cidr_blocks = ["0.0.0.0/0"]
25  }
26
27  tags = {
28    Name = "app-sg"
29  }
30 }
```

7- create ec2(bastion) in public subnet with security group from 7

The screenshot shows the Visual Studio Code interface with a Terraform project. The Explorer panel on the left lists files under a 'TERRAFORM' directory, including 'Day1' and various .tf files. The 'data.tf' file is selected and its content is displayed in the main editor. The code defines a data source for an AWS AMI, filtering for the most recent AMI owned by 'amazon' with the name 'amzn2-ami-hvm-*x86_64-gp2'.

```
1 data "aws_ami" "amazon_linux" {
2   most_recent = true
3   owners      = ["amazon"]
4
5   filter {
6     name     = "name"
7     values   = ["amzn2-ami-hvm-*x86_64-gp2"]
8   }
9 }
```

```
Terminal  Help  ec2-bastion.tf - Terraform - Visual Studio Code
ite.tf  route-associations.tf  security-group-bastion.tf  security-group-app.tf  ec2-basti

Day1 > ec2-bastion.tf
1  resource "aws_instance" "bastion" {
2      ami = data.aws_ami.amazon_linux.id
3      instance_type = "t2.micro"
4
5      subnet_id          = aws_subnet.public_1.id
6      vpc_security_group_ids = [aws_security_group.bastion_sg.id]
7
8      associate_public_ip_address = true
9
10     tags = {
11         Name = "bastion-ec2"
12     }
13 }
```

8- create ec2(application) private subnet with security group from 8

```
Terminal  Help  ec2-app.tf - Terraform - Visual Studio Code
ite.tf  route-associations.tf  security-group-bastion.tf  security-group-app.tf  ec2-basti

Day1 > ec2-app.tf
1  resource "aws_instance" "app" {
2      ami = data.aws_ami.amazon_linux.id
3      instance_type = "t2.micro"
4
5      subnet_id          = aws_subnet.private_1.id
6      vpc_security_group_ids = [aws_security_group.app_sg.id]
7
8      associate_public_ip_address = false
9
10     tags = {
11         Name = "app-ec2"
12     }
13 }
```

File Edit Selection View Go Run Terminal Help

data.tf - Terraform - Visual Studio Code

EXPLORER

- TERRAFORM
 - Day1
 - .terraform
 - .terraform.lock.hcl
 - data.tf
 - ec2-app.tf
 - ec2-bastion.tf
 - internet-gateway.tf
 - provider.tf
 - route-associations.tf
 - route-table-private.tf
 - route-table-public.tf
 - routes.tf
 - security-group-app.tf
 - security-group-bastion.tf
 - subnet-private.tf
 - subnet-public.tf
 - terraform.tfstate
 - terraform.tfstate.backup
 - vpc.tf

Day1 > data.tf

```
1 data "aws_ami" "amazon_linux" {
2   most_recent = true
3   owners      = ["amazon"]
4
5   filter {
6     name   = "name"
7     values = ["amzn2-ami-hvm-*x86_64-gp2"]
8   }
9 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL JUPYTER

terraform

```
aws_instance.bastion: Creating...
aws_instance.app: Creating...
aws_instance.bastion: Still creating... [00m10s elapsed]
aws_instance.app: Still creating... [00m10s elapsed]
aws_instance.bastion: Still creating... [00m20s elapsed]
aws_instance.app: Still creating... [00m20s elapsed]
aws_instance.bastion: Still creating... [00m30s elapsed]
aws_instance.app: Still creating... [00m30s elapsed]
aws_instance.bastion: Still creating... [00m40s elapsed]
aws_instance.app: Still creating... [00m40s elapsed]
aws_instance.bastion: Creation complete after 46s [id=i-09aad08b670f58e9c]
aws_instance.app: Creation complete after 47s [id=i-0e1210921d701f2ff]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
PS C:\Users\lascom\Desktop\ititask\Terraform\Day1>
```

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#vpcs

Search [Alt+S]

United States (N. Virginia)

Rana salem (656019697915)

Rana salem

VPC > Your VPCs

VPC dashboard <

AWS Global View

Filter by VPC

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Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Your VPCs

VPCs VPC encryption controls

Your VPCs (1/3) Info

Find VPCs by attribute or tag

1

main-vpc vpc-0f237af29007dcab2 Available - -

vpc-0f237af29007dcab2 / main-vpc

DNS resolution Enabled	Tenancy default	DHCP option set dopt-0a166477954da5cb2	Main route table rtb-0e2bed38668856bcf
Main network ACL acl-0f948fc9c9c0b08a3	Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 656019697915
Encryption control ID -	Encryption control mode -		

aws

Search

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United States (N. Virginia)

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VPC > Internet gateways > igw-0862922de98a5d179

igw-0862922de98a5d179 / main-igw

Actions

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Details

Info

Internet gateway ID

igw-0862922de98a5d179

State

Attached

VPC ID

vpc-0f237af29007dcab2 | main-vpc

Owner

656019697915

Tags (1)

Manage tags

Search tags

Key	Value
Name	main-igw

aws

Search

[Alt+S]

Ask Amazon

United States (N. Virginia)

Rana salem (656019697915)

VPC > Route tables > rtb-0b53dc398c67ed1ee

rtb-0b53dc398c67ed1ee / public-rt

Actions

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Info

Route table ID

rtb-0b53dc398c67ed1ee

Main

No

Explicit subnet associations

2 subnets

Edge associations

-

VPC

vpc-0f237af29007dcab2 | main-vpc

Owner ID

656019697915

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0862922de98a5d179	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

aws

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Ask Amazon

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VPC > Route tables > rtb-0ff5692ebd0f58662

rtb-0ff5692ebd0f58662 / private-rt

Actions

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Details

Info

Route table ID

rtb-0ff5692ebd0f58662

Main

No

Explicit subnet associations

2 subnets

Edge associations

-

VPC

vpc-0f237af29007dcab2 | main-vpc

Owner ID

656019697915

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

VPC dashboard

AWS Global View

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 - Route covers

subnet-048b49314bd0cd97d / public-subnet-1

Details

Subnet ID subnet-048b49314bd0cd97d	Subnet ARN arn:aws:ec2:us-east-1:656019697915::subnet/subnet-048b49314bd0cd97d	State Available	Block Public Access ☐ Off
IPv4 CIDR 10.0.1.0/24	Available IPv4 addresses 250	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone us-east-1a (us-east-1)	Network border group us-east-1	VPC vpc-0f237af29007dcab2 main-vpc	Route table rtb-0b53dc398c67ed1ee public-rt
Network ACL acl-0f948fc9c9c0b08a3	Default subnet No	Auto-assign public IPv4 address Yes	Auto-assign IPv6 address No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -	Outpost ID -	IPv4 CIDR reservations -
IPv6 CIDR reservations -	IPv6-only No	Hostname type IP name	Resource name DNS A record Disabled
Resource name DNS AAAA record Disabled	DNS64 Disabled	Owner 656019697915	

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) Rana salem (656019697915) Rana salem

VPC > Subnets > subnet-01f5865b365eb951e

VPC dashboard

AWS Global View [↗](#)

Filter by VPC ▾

▼ Virtual private cloud

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- Subnets**
- Route tables
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- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

subnet-01f5865b365eb951e / private-subnet-2

Actions ▾

Details

Subnet ID subnet-01f5865b365eb951e	Subnet ARN arn:aws:ec2:us-east-1:656019697915:subnet/subnet-01f5865b365eb951e	State Available	Block Public Access Off
IPv4 CIDR 10.0.4.0/24	Available IPv4 addresses 251	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone use-1-az2 (us-east-1b)	Network border group us-east-1	VPC vpc-0f237af29007dcab2 main-vpc	Route table rtb-0ff5692ebd0f58662 private-rt
Network ACL acl-0f948fc9c9c0b08a3	Default subnet No	Auto-assign public IPv4 address No	Auto-assign IPv6 address No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -	Outpost ID -	IPv4 CIDR reservations -
IPv6 CIDR reservations -	IPv6-only No	Hostname type IP name	Resource name DNS A record Disabled
Resource name DNS AAAA record Disabled	DNS64 Disabled	Owner 656019697915	

Actions

Actions

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VPC > Security Groups

Security Groups (1/11) Info

Find security groups by attribute or tag

Actions ▾ Export security groups to CSV ▾ Create security group

	Name	Security group ID	Security group name	VPC ID
☐	-	sg-0191e940b4858bd31	default	vpc-0267cc116fc2ad514
☒	bastion-sg	sg-01e232cc751004b36	terraform-2026050112424111420000...	vpc-0f237af29007dcab2
☐	-	sg-0543bc00e91da311e	default	vpc-0f237af29007dcab2

sg-01e232cc751004b36 - terraform-20260501124241114200000004

Inbound rules (1)

Search

Manage tags Edit inbound rules

	Name	Security group rule ID	IP version	Type	Protocol
☐	-	sgr-07d0a03797fe6dbb9	IPv4	SSH	TCP

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VPC > Security Groups

Security Groups (1/11) Info

Find security groups by attribute or tag

Actions ▾ Export security groups to CSV ▾ Create security group

	Name	Security group ID	Security group name	VPC ID
☐	-	sg-081a64a4ec9633182	ec2_sg	vpc-0267cc116fc2ad514
☐	-	sg-07356b07c041c9063	launch-wizard-2	vpc-00b61b98dc35e2a23
☒	app-sg	sg-0fb94de79a5f53cea	terraform-2026050112424111390000...	vpc-0f237af29007dcab2
☐	-	sg-03214dd45a2df4845	launch-wizard-3	vpc-00b61b98dc35e2a23

sg-0fb94de79a5f53cea - terraform-20260501124241113900000003

Inbound rules (2)

Search

Manage tags Edit inbound rules

group rule ID	IP version	Type	Protocol	Port range	Source
ead6b58be8ab8	IPv4	Custom TCP	TCP	3000	10.0.0.0/16
de5a2b0d906da	IPv4	SSH	TCP	22	10.0.0.0/16

aws

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United States (N. Virginia)

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EC2 > Instances

EC2

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Instances (2) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch Instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
☐	app-ec2	i-0e1210921d701f2ff	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a
☐	bastion-ec2	i-09aad08b670f58e9c	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a

Select an instance